PUSL3190 Computing Project

Project Proposal

Cre8tify Web Application

Supervisor: Miss Dharani Rajasinghe

Name: Mahara Mahara
Plymouth Index Number: 10952728
Degree Program: BSc (Hons) Software Engineering

Contents

Chapter 01: Problem statement 3

Chapter 02: Project description 4

2.1. Project Overview 4

2.2. System user and their roles 4

2.3. Background and Rationale 5

2.4. Objectives 5

2.5. Scope 6

2.6. Expected Outcomes 6

2.7 Project keywords 6

Chapter 03: Research Gap 7

Chapter 04: Requirements Analysis 8

Chapter 05: Finance 9

Chapter 06: External Organizations 10

Chapter 07: Time Frame / Timeline 10

Referencing / Bibliography 11

Table of tables

Table 1: Requirement Analysis 9

Table 2: Finance 9

Table of figures

Figure 1: Gantt Chart 10

Chapter 01: Problem statement

This project addresses the lack of localized platforms that allows both Sri Lankan designers and buyers to collaborate within a single platform for custom t-shirt designing and selling. In the modern era, most of the youth and young adults pay more concerns on their outfits by wanting to have unique designs for them. In this case, many tends to follow online custom design platforms which make it easier for them to match with their vibe and customize as they wish. On the other hand, there is also a set of young, creative minds who are interested on designing and wanting to sell their designs/products to the market. But existing platforms like "Printify" doesn't allow free service from start to end. Also, when it comes to building a market place, the local beginners may find it difficult since platforms like "Printify" focus on global market and they rely on international payment methods like PayPal which is not fully supported in Sri Lanka.

Therefore, local designers face difficulties in finding free platforms to design and sell their design/products. On the otherside, most of the buyers do not get to virtually fit on their clothes before purchasing. Therefore, due to lack of customization options, local payment methods and interactive previews, it creates a gap between the creativity and accessibility. As a result, by building a locally adaptable custom t-shirt designing platform, we can help both the young designers and design enthusiastic buyers. Alongside, we can bring Sri Lanka's e-commerce industry into another extent while also awakening the hidden creative minds.

Therefore, the main problem addressed in this project is the lack of localized, user- friendly, and affordable platform where we can design, preview, and buy custom design t-shirts while allowing designers across our country to showcase and sell their designs for an affordable price within Sri Lanka.

As an undergraduate interested in digital designing, I have personally faced difficulties when it comes to designing, creating your own marketplace and selling. That's when I found "Printify" which has a various collection for designing, but as a Sri Lankan beginner it wasn't easy at all to build a marketplace using Etsy and to receive payments using PayPal. Experiencing these made me realize that many other young designers in Sri Lanka face the same limitations and it inspired me to come up with "Cre8tify", a platform for the beginner

designers around our country to freely create their imaginations into unique t-shirt designs and comfortably earn within a local system.

Chapter 02: Project description

2.1. Project Overview

"Cre8tify" is a web based, mobile responsive, e-commerce platform designed for purchasing and custom designing t-shirts. This application connects users, who creates designs on our platform for selling with the users, who will be purchasing or customizing those designs. What makes our platform outstand is, unlike other digital designing platforms, we give hands for our beginner designers with simple, user-friendly design experience. Not only the designers, even users can customize an existing design they like and try them out with live design preview option before purchasing. We will also implement local payment methods by solely focusing on Sri Lankan users.

2.2. System user and their roles

Users (Buyers)

- Log into user dashboard and browse through the existing t-shirt designs.
- Try live preview (only front) by uploading your image or a dummy model.
- Request designer to edit as preferred.
- Customize by your own (color, text, small edits).
- Give likes on the existing designs.
- Add to cart and select a payment method (bank transfer/sandbox)
- Download approved digital files if you have purchased only for the design, not the whole product.

Designers

- Log into designer dashboard.
- Upload or create new designs using the toolbox given.
- Set the design price and description and submit for admin approval.
- Preview, edit or check the status(draft/submitted/approved/rejected) of the designs created.
- Do changes on their existing designs requested by the user.
- Track on the sales and earnings.

Admin

- Log into admin dashboard.
- Manage user and designer accounts.
- Approve and reject submitted designs.
- Verify order payments and maintain records.
- Review reported problems and take necessary actions.
- Adjust the price of original t-shirt and service charge.

2.3. Background and Rationale

The need of creating "Cre8tify" arose due to the rising interest of online designing platforms. Even though, platforms like Printify exists, they only focus on global designers. Therefore, local designers face difficulties due to complexity in setting up their shops, affordability and limited payment options. As a result, this project aims on providing a simplified platform that handles the entire process from designing to purchasing within a single platform.

2.4. Objectives

- Developing a localized ecommerce platform for Sri Lankan users to custom design t-shirts and purchase.
- Introducing a live preview option for users to fit in a collection of designs on our website before purchasing.
- Providing a free space for local designers to design on their work easily.
- Allowing local payment options to make transactions flow smoother.
- Establishing admin control to maintain platform quality by approving designs and verifying payments.
- Delivering a responsive design for smooth accessibility across desktop, laptop and mobile screens.

2.5. Scope

Cre8tify's scope is on T-shirt designing and purchasing them for men, women and kids. Our system includes three main user roles such as, user (buyer), designer and admin. Here, users can browse, preview, customize and purchase designs or the full t-shirt. Designers can create designs using our tool kit or simply upload designs, set prices and manage their portfolio. Admin is responsible for handling and managing accounts, design approvals and payment verification. For users, we'll be implementing dual preview options, live preview (limited to front-view) where user uploads their image and fit in with the t-shirt design they chose which is a unique feature in our application. And the other option is a dummy preview (3D model) allowing a 360° rotation displaying the design from all the sides. These will be handled ethically protecting users' privacy. Unlike Printify, Cre8tify uses local payment options (bank transfer/Sandbox) and is a web based, mobile responsive application.

2.6. Expected Outcomes

The expected outcome of this project is a fully functional, user friendly, and responsive web application allowing designers to design and sell their products and users to customize, preview and purchase t-shirts online with local payment methods. The final system prototype will present live design previews, secure payment integration and proper role-based access for designers, users, and admin by contributing to **Sri Lanka's digital designing and e-commerce** ecosystem.

2.7 Project keywords

Keywords: E-commerce, Custom T-shirt Design, Local Payment Gateway, Responsive Web Application, User–Designer Collaboration.

Chapter 03: Research Gap

Due to the rising interest on digital designing, there are many online custom designing platforms like “Printify”, “Teespring”, “Redbubbles” but they focus mainly on global market connecting with global print providers etc. According to Printify (2025), this platform allows designers to build up their own stores with online marketplaces like Etsy and Shopify, enabling to sell printed items worldwide. Also, Redbubble (2024) provides designers to upload their artworks and sell printed t-shirts and accessories to the global market.
However, most of these platforms supports international markets and depends on global payment methods like PayPal and Stripe which really don’t work well in Sri Lanka. Also, this might be difficult for the beginner designers and students due to subscription fees and maintaining third-party integrations. These platforms lack options like live preview using customer’s image or 360° dummy preview. And it doesn’t allow localized communication between the user and the designer as well. Users only get to purchase the existing product from the store.

Although MYCloth, an experimental research project, explores intelligent and interactive T-shirt customization based on user preferences (Liu & Wang, 2024), such models remain limited to research and are not publicly available or localized for small-scale creators.
As a result, there’s a clear need for a localized, free-affordable, user friendly custom t-shirt designing system for Sri Lankan users. Cre8tify will be able to connect the gap between the designers and users while also implementing localized payment options. Not only them but introduce users with the live preview and 360° dummy preview options where they can virtually fit on t-shirts before purchasing. All of these features are new and are not currently combined in any existing system. With the implementation of Cre8tify, it may lead to uplift the e-commerce sector of Sri Lanka as well as help beginner designers to earn an income.

Chapter 04: Requirements Analysis

Type	**Description**
Functional Requirements	• The system shall allow users to register and login securely, browse through existing designs, preview, customize and purchase t-shirts/designs. • The system shall allow designers to create and upload designs with free tools on our website. • The system shall allow admin to filter designers and users when logged in and approve and reject designs submitted by designers. • The system shall provide a live preview option and 360° dummy preview option for the users to visualize the t-shirts on them.
Non-functional Requirements	• The system should load pages within 5 seconds. • The system should ensure secure login and data protection. • It should be mobile responsive and be able to handle up-to 100 concurrent users without crashing. • Interface should be simple, user friendly and visually appealing.

User Requirements	• Users should be able to login, browse t-shirts, preview, customize and purchase them. • Designers should be able to edit, upload, set price and manage their designs and shop freely. • Admin should be able manage both users and designers, verify payments, manage accounts and monitor their activities.

Table 1: Requirement Analysis

Chapter 05: Finance

Item	Description	Estimated Cost (LKR)
Personal Laptop	Used for system designing, development and testing	0.00
Mobile Phone	For testing mobile responsiveness	0.00
Figma/Visual Studio Code/ GitHub/ Node.js/ MySQL	All the software tools used are open source	0.00
Internet connection	For research, development, and uploading to GitHub	2,500.00
Printing and documentation	Reports and submission materials	500.00
Optional hosting	For the final demo/ deployment	1,000.00

Table 2:Finance

- **Total estimated cost = LKR 4,000.**

Note: all the development tools used here are free/ open source. The above cost is depending on documentation, internet, and hosting.

Chapter 06: External Organizations

This project is an individual academic work carried out by myself under the guidance of the assigned supervisor, Miss Dharani Rajasinghe from NSBM Green University affiliated with Plymouth University. In this project there's no external companies or institutions directly involved. During the backend development, data handling or system handling, I might seek for guidance from our lecturers and the lab staff. Also, I might ask my peers for feedback regarding the UI/UX testing.

Chapter 07: Time Fame / Timeline

1	Task/Activity	Start Date	End Date	September	October	November	December	January	February	March	April
2	Topic selection and supervisor approval	9/10/2025	9/16/2025								
3	Requirement gathering and flow diagrams	9/17/2025	9/30/2025								
4	UI/UX design (figma prototypes)	10/1/2025	10/25/2025								
5	Proposal submission	10/31/2025	10/31/2025								
6	Frontend development	11/1/2025	12/15/2025								
7	Backend development and database setup	1/1/2026	1/31/2026								
8	Sytem integration and payment module	2/1/2026	2/20/2026								
9	Testing and debugging	3/1/2026	3/20/2026								
10	Final documentation and presentation	3/21/2026	4/15/2026								

Figure 1: Gantt Chart showing the proposed project timeline for Cre8tify from September 2025 to April 2026.

Referencing / Bibliography

- **Printify**
  In-text: (Printify, 2025)
  Reference list: Printify (2025) *Print on Demand – Custom Apparel*. Available at: https://printify.com (Accessed: 12 October 2025).

- **Redbubble**
  In-text: (Redbubble, 2024)
  Reference list: Redbubble (2024) *Make Your Own T-Shirts for Sale*. Available at: https://www.redbubble.com (Accessed: 12 October 2025).

- **Liu and Wang (MYCloth research paper)**
  In-text: (Liu and Wang, 2024)
  Reference list: Liu, Y. and Wang, L. (2024) *MYCloth: Towards Intelligent and Interactive Online T-Shirt Customization*. Available at: https://arxiv.org/abs/2404.15801 (Accessed: 12 October 2025).

- **Özbek (2020)**
  In-text: (Özbek, 2020)
  Reference list: Özbek, A. (2020) *Online customized T-shirt design and evaluation of online websites for customization*. *Industria Textila*, 71(4), pp. 371–379. Available at: https://www.revistaindustriatextila.ro/images/2020/4/011%20AHMET%20%C3%96ZBEK_Industria%20Textila%204_2020.pdf (Accessed: 12 October 2025).

- **Kazi et al. (2021)**
  In-text: (Kazi *et al.*, 2021)
  Reference list: Kazi, S., Singh, A., Kumar, M. and Mhatre, M. (2021) *Online T-Shirt Designing*. *International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE)*. DOI: 10.17148/IJARCCE.2021.10565. Available at: https://ijarcce.com/papers/online-t-shirt-designing/ (Accessed: 12 October 2025).